

Ahmed Ikram | Software Engineering

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Final-year BSc Computer Science (Data Science & AI) student at the University of Dundee (expected First-Class, 2027). Experienced in building production systems end-to-end, picking the pattern that fits the load: a **Rust** extension where Python was the bottleneck, ARM64 where it cuts compute cost roughly 20-30% at equal performance. An event-driven backend and a full-stack real-time platform, both OIDC-deployed with 1,400+ tests apiece; external APIs are streamed, rate limited, and behind cheaper fallbacks. Seeking a **Software Engineering** role.

EDUCATION

University of Dundee - BSc (Hons) Computer Science (Data Science and AI)

Sep 2023 - Present

- On track for a **First-Class Honours** degree. Expected Graduation: **June 2027**

TECHNICAL SKILLS

Languages: Python · TypeScript · JavaScript · SQL · T-SQL · Rust · Bash

Frontend: React · React Native · Expo · Tailwind CSS · Vite

Backend & APIs: Flask · FastAPI · Gunicorn · Pydantic · SQLAlchemy · Apache Kafka · Nginx

Security: JWT · OAuth 2.0 · RBAC · bcrypt · AWS Secrets Manager

Databases: PostgreSQL · Redis · SQLite · Alembic

Cloud & DevOps: AWS (ECS Fargate, EC2, S3, RDS, ElastiCache, WAF, IAM, CloudFront, Secrets Manager) · Azure (Databricks, Azure SQL, ADLS) · Terraform · Docker · GitHub Actions · OIDC · Linux · Socket.IO

Observability & Testing: Grafana · Prometheus · StatsD · Pytest · Vitest · Jest · Playwright · Cypress · React Testing Library · mutmut · Hypothesis

PROJECTS

DevSync Project Management Platform with GitHub Integration

React · Tailwind · Flask · PostgreSQL · Socket.IO · Gunicorn · GitHub OAuth · AWS · Docker · GitHub Actions

- Full-stack collaboration platform: Flask REST API over a 12-table indexed PostgreSQL schema, and a JWT-authenticated Socket.IO layer broadcasting live task updates and dashboard refreshes with zero cross-project leakage; Gunicorn's gevent worker lets WebSocket and HTTP traffic share one async-I/O process without async-ifying the whole codebase.
- Three-tier RBAC via route decorators, server-side GitHub OAuth CSRF validation and encrypted tokens; bidirectional Issue/PR ↔ task sync keeping DevSync and GitHub state consistent in both directions; dual auth (HTTP-only cookie + bearer) so the same JWT works for the browser and API clients.
- AWS** three-tier VPC (CloudFront S3/SPA → ALB → ECS Fargate → private RDS) with OIDC-federated CI/CD and zero-long-lived credentials; multi-stage Docker build cut the runtime image 45% (600MB → 330MB) holding sub-300ms p99 latency. CD waits for ECS health checks before promoting the front and backend.
- Guarded by 1,452 automated tests (Pytest, Jest + Cypress) at an 85%/85% coverage gate, with integration tests against PostgreSQL and path-aware CI; CodeQL security scanning and dependency auditing reject any PR that drops coverage or fails a check before it reaches deployment.

ATM Log Aggregation, Anomaly Detection & Diagnostics Platform

FastAPI · React · PostgreSQL · Kafka · Redis · Docker · Terraform · AWS

- Backend engineering lead on a 7-person Agile team for NCR Atleos: independently architected a 13-service Dockerised platform, full AWS production deployment, and a 3-pipeline CI/CD system as a post-deadline extension on the team-delivered foundation.
- FastAPI REST API (30 endpoints, 6 routers, JWT + Redis-blacklist auth) with health-check-cascading startup across Kafka, Redis, PostgreSQL; Kafka Raft consumer gives at-least-once delivery with hybrid Redis/LRU deduplication and a Redis Stream DLQ, holding 100 msg/s over 2.5M events.
- 8 Redis patterns (rate-limit, dedup, lock, Pub/Sub, cache, JWT blacklist, DLQ, counters) share one connection pool and degrade independently; JSONB schema-on-read means a new log source is a parser, not a migration; React frontend ships as a ~25MB nginx-alpine image.
- Deployed via Terraform across a multi-AZ VPC (2 public + 2 private subnets), ECS Fargate behind an ALB, an EC2 host running Kafka and Redis in a private subnet.
- 3-pipeline CI/CD system: CI runs 1,438 tests (959 pytest, 394 vitest, 10 Playwright E2E, 75 Terraform assertions) and Checkov IaC compliance on every PR, then rolls out zero-downtime via OIDC with zero long-lived credentials.

StockLens FinTech Spending & Investment App

React Native · Expo · FastAPI · PostgreSQL · Redis · Rust · Terraform · Docker · AWS

- **Async** FastAPI backend: 59 REST API endpoints across 14 routers with **biometric-backed JWT** auth, **bcrypt** hashing, per-route rate limiting; **Agent** assistant endpoint is SSE-streamed with retry/backoff and a two-tier Redis + PostgreSQL session store that survives process restarts.
- Native **Rust** extension (PyO3/Maturin, 12 functions) replaces a pandas bottleneck in technical-indicator computation with zero-cost abstractions, cross-built for ARM64 via QEMU in CI and called by both the LSTM training pipeline and live API.
- Deployed via Terraform (14 modules, 193 resources): three-tier network isolation (WAF-protected ALB → ECS Fargate ARM64 → private RDS/Redis) chosen for a 20-30% compute-cost cut over x86 at equal performance, with CPU/RPS autoscaling; multi-stage images are 450MB vs 1.2GB naive.
- 1,415 backend and 822 frontend tests at a 90%/90% coverage gate, including property-based tests on financial invariants and nightly mutation testing on critical paths; enforced by a 9-job CI and a 7-stage OIDC-authenticated CD pipeline.

W3C Web Logs ETL Pipeline

Airflow · Databricks · dbt · PySpark · Azure SQL · Terraform · Power BI · Grafana · Prometheus · Marquez · OpenLineage

- Databricks DLT medallion pipeline ingests 93 W3C IIS log files and enriches 153,377 Silver rows before export to Azure SQL; 7 DLT quality gates reject malformed records with 0 rows dropped.
- 16 dbt models compile against PostgreSQL and T-SQL from one codebase via 18 compatibility macros; 121 dbt tests gate every model, catching 213 duplicate rows through a 12-component deduplication key.
- Integrated OpenLineage with a self-hosted Marquez collector across all 4 Airflow DAGs, combining task-run events, dbt model/column lineage, Azure SQL dataset edges, and custom data-quality facets into one cross-engine lineage graph.
- Deployed 30+ Azure resources through Terraform and OIDC federation; 620 tests, a freshness probe, 3 Grafana dashboards, and 10 alert rules validate infrastructure, pipeline health, and data quality.

CERTIFICATES

<u>AWS Academy Graduate - Machine Learning Foundations</u>	<i>Feb 2025</i>
<u>Foundations of Azure AI: Concepts, Capabilities, and Implementation</u>	<i>Oct 2024</i>
<u>AWS Academy Graduate - Cloud Foundations</u>	<i>Mar 2024</i>

EXPERIENCE

Beaully Mini Market , Dundee, Scotland	
<i>Retail Assistant</i>	<i>September 2023 - Present</i>
Zulus Peri Peri Grill , Dundee, Scotland	
<i>Front of House Worker</i>	<i>June 2021 - September 2023</i>